5) How many ways 3 ranks in 3 subjects to 20 students?

Ans:-

$$\begin{array}{cccc} & M & P & C \\ 20 & \underline{1^{st}} & \underline{1^{st}} & \underline{1^{st}} \\ 19 & \underline{2^{nd}} & \underline{2^{nd}} & \underline{2^{nd}} \\ 18 & \underline{3^{rd}} & \underline{3^{rd}} & \underline{3^{rd}} \\ & \downarrow & & \\ & 20P_3 & 20P_3 & 20P_3 \end{array} = (20P_3)^3 = \underline{\underline{(6840)^3}}$$

6) How many ways 20 ranks in 3 subjects to 20 students?

$$\begin{array}{ccccc} \text{Subjects} & & M & P & C \\ & & - & - & - \\ & & - & - & - \\ \text{ranks} & \left\{ \right. & - & \vdots & - \\ & & \vdots & \vdots & \vdots \\ & & 20 & 20 & 20 \\ & & 20P_{20} & 20! & 20! \\ & & =20! & & \end{array} \qquad 20! \times 20! \times 20! = \underline{\underline{(20!)^3}}$$

7) How many ways 20 ranks in 20 subjects to 20 students?

$$\begin{array}{cccccccccc} \text{Subjects} & M & P & C & - & - & - & - & - & 20 \text{ subjects} \\ & - & - & - & - & - & - & - & & \\ 20 \text{ ranks} \left\{ \right. & - & - & - & , & , & & & & = \underline{\underline{(20!)^{20}}} \\ & \vdots & \vdots & \vdots & ' & ' & ' & ' & & \\ & 20! & 20! & 20! & - & - & - & - & & 20 \text{ times} \end{array}$$

8) How many 3 digit number plates can be formed.

(i) with 4 digits 1, 2, 3 & 4?

Ans:- default w/o repetition we have to consider.

$\Rightarrow$ $nP_r \Rightarrow n=4, r=3 \Rightarrow 4P_3 = 4\times3\times2 = 24$ plates.

(ii) which are even numbers with 4 digits 1, 2, 3 & 4.

$$\underbrace{\underline{}\;\underline{}}_{3P_2}\;\overset{2 \text{ ways}}{\underline{}}_{2 \text{ or } 4} = 3! \times 2 = \text{~~6 plates only~~}.$$

$= 6\times2 = 12$ plates.

(iii) which are 5 multiples with 6 digits 0,1,2,3 & 4 & 5?

Ans:- $\underbrace{\_\ \_}_{5P_2}\ \underset{\substack{5\,(or)\,0 \\ 1\ digit}}{\overset{2\ ways}{\_}}$ [Note:- It's a number plate. So, '0' is also considered].

$\Rightarrow 5P_2 \times 2 = \frac{5!}{(5-2)!} \times 2 = \frac{5!}{3!} \times 2 = 5 \times 4 \times 2 = 40$ plates.

(iv) which are 4 multiples with digits 1,2,4,8,3?

$\underset{\substack{3P_1 \\ =3}}{\_}\ \_\ \_$ [to become 4 multiple last 2 digits divisible by 4.]

1 2
2 4.
2 8
3 2
4 8
8 4.

6 ways [w/o repetition default that's why 44 288 ~~should~~ not considered.]

$\Rightarrow 3 \times 6 = 18$ plates.

⑨ How many 5 digited numbers with digits 0,2,3,4,5?

$\_\ \_\ \_\ \_\ \_$ [here '0' is not considered in the beginning to make 5 digited number]

$(2,3,4,5) \rightarrow 4\ \underbrace{4\ 3\ 2\ 1}_{4P_4 = 4!}$

$4^2 \times 6 = 96$ numbers

$= 4 \times 4! = \overline{4 \times 24} = 96$

(10) How many words with word 'vectorindia'?

$= \frac{11!}{2!\,(2\ i's)} = 11 \times 10 \times 9 \times 8 \times 7 \times 6 \times 5 \times 4 \times 3 = 1,99,58,400$

(11) Find the no. of different words that can be formed from the word 'success'?

$= \frac{7!}{3!\,(3\ s's) \cdot 2!\,(2\ c's)} = \frac{7!}{3 \times 2 \times 2} = \frac{5040}{12}$

$= 420$ (or) $\frac{7 \times 6 \times 5 \times \overset{2}{\cancel{4}} \times \cancel{3} \times \cancel{2} \times 1}{\cancel{3} \times \cancel{2} \times \cancel{2}} = \underline{420}$

⑫ How many even numbers of 3 digits can be formed. with the digits 0, 1, 2, 3, 4, 5 & 6?

Ans:- here we have to consider ending with zero & does-not end with zero.

case(i):- end with zero:-

$\underset{5\text{ways}}{\underset{\downarrow}{\_}}\ \ \underset{6\text{ways}\,(1,2,3,4,5,6)}{\underset{\downarrow}{\_}}\ \ \overset{0}{\underset{\downarrow\text{fixed in 'i' way}}{\_}}$ = 5×6×1 = 30 arrangements

case(ii):- does not end with zero:-

$\underline{5}\ \ \underline{5}\ \ \underset{2/4/6}{\underline{3\text{ ways}}}$

[zero doesnot come & already one digit placed in units place so.
7−(2 ⇒ 0 & 1 units digit) = 5 ways]

→ two digits placed @ units & hundreds positions then remaining 5 digits can be placed including zero in one place in 5 ways.

⇒ 5×5×3 = 75 arrangements.

Total = 30+75 = 105 3 digited even numbers.

⑬ A round table conference is to be held b/w delegates of 15 companies. In how many ways can they be seated if delegates from two MNC's may wish to sit together?

Ans:- from 2 multinational company delegates together means we have to consider them as one & arrange.